Email: taneemishere@gmail.com
Website: taneemishere.github.io
LinkedIn: linkedin.com/in/taneemishere

Taneem Ullah Jan

Applied AI researcher and engineer building reliable, production-grade systems across LLMs, agentic workflows, multimodal learning, and generative modeling. Experienced in translating research into deployed systems through architecture, retrieval, memory, evaluation, orchestration, and cloud infrastructure.

Professional Experience

AI Lead
Oct. 2025 – Present

ErisAI

- Lead the research, architecture, and engineering of production-grade AI systems spanning LLMs, multimodal models, retrieval, memory, and agentic workflows.
- Architect and develop Synapse, a research-driven AI infrastructure layer integrating hybrid retrieval, multi-turn memory, intelligent routing, and model orchestration for reliable real-world deployments.
- Define evaluation protocols and operational controls for response quality, traceability, knowledge access, and failure analysis across enterprise AI systems.

AI Researcher
Sep. 2024 – Oct. 2025

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models; the system was adopted by fashion brands and increased user engagement by 40%. (Demo)
- Developed controllable 2D and 3D digital human systems with pose estimation and generative modeling, supporting personalized experiences and reducing product return rates by 30%.
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following in customer support and interactive applications.

Research AI Engineer
Jan. 2023 – Dec. 2023

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. (Use cases)
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Intern Machine Learning Engineer
Aug. 2021 – Nov. 2021

NAECO Blue GmbH [*Profile*]

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

Publications

Taneem, U. J., & Ayesha, N. (2024). Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs. International Journal of Innovations in Science & Technology, (IJIST) 6(5), 165–179, Special Issue ICTIS 2024.

Taneem, U. J., & Zakira, I. (2022). HTML Code Generation from Images with Deep Neural Networks. Journal of Engineering and Applied Sciences (JEAS), UET Peshawar. **(Young Undergraduate Researcher Award)**

Selected Research Projects

Synapse (2026). Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.

DarwinPatch (2026). Verifier-grounded coding-agent repair framework combining bounded failure evidence, route-aware search, regression gates, and auditable benchmark evaluation.

DGM-LLM (2025). Autonomous code self-improvement framework combining evolutionary search with LLM-guided mutation and multi-dimensional evaluation.

VMVLM (2025). Vision-language architecture using direct visual feature modulation to improve multimodal reasoning and instruction-following.

EmbedVoiceLLM (2025). Efficient multimodal framework mapping speech representations directly into LLM embedding space for low-latency voice interaction.

FlexiSMPL (2025). Real-time SMPL body modeling system with measurement-driven control and interactive 3D visualization.

Skills

- **Programming**: Python, C++, Bash, MATLAB
- **ML / AI**: PyTorch, Transformers, TensorFlow, NumPy, fine-tuning, multimodal learning, model evaluation, experiment design
- **AI Systems**: Agentic workflows, retrieval-augmented generation, hybrid retrieval, LLM orchestration, LangChain
- **Computer Vision / 3D**: OpenCV, PyTorch3D, MediaPipe, ONNX
- **Infrastructure**: Docker, Git, Linux, Weights & Biases, AWS, Azure, GCP

Education

University of Engineering and Technology Peshawar, Pakistan

Bachelor of Science in Computer Science, CGPA: 3.58/4.0 — Sep. 2018 – Sep. 2022